

CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ SOLUTIONS

Chapter 1: Real Numbers

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 HCF and LCM by prime factorisation

1. If two positive integers p and q can be expressed as $p = 18a^2b^4$ and $q = 20a^3b^2$, where a and b are prime numbers, then LCM (p, q) is: (a) $2a^2b^2$ (b) $180a^2b^2$ (c) $12a^2b^2$ (d) $180a^3b^4$ **CBSE 2024, 1 mark**

SOLUTION

$$p = 2 \times 3^2 \times a^2 \times b^4 \text{ and } q = 2^2 \times 5 \times a^3 \times b^2$$

LCM takes the greatest power of every prime: $2^2 \times 3^2 \times 5 \times a^3 \times b^4 = 180a^3b^4$

Answer: (d)

2. In a teachers' workshop, the number of teachers teaching French, Hindi and English are 48, 80 and 144 respectively. Find the minimum number of rooms required if in each room the same number of teachers are seated and all of them are of the same subject. **CBSE 2024, 3 marks**

SOLUTION

For the fewest rooms, each room must seat the greatest possible number, which is the HCF of 48, 80 and 144.

$$48 = 2^4 \times 3, 80 = 2^4 \times 5, 144 = 2^4 \times 3^2, \text{ so HCF} = 2^4 = 16$$

$$\text{Number of rooms} = \frac{48}{16} + \frac{80}{16} + \frac{144}{16} = 3 + 5 + 9$$

Answer: 17 rooms

2 The Fundamental Theorem of Arithmetic

3. Show that the number $5 \times 11 \times 17 + 3 \times 11$ is a composite number. **CBSE 2024, 2 marks**

SOLUTION

$$5 \times 11 \times 17 + 3 \times 11 = 11(5 \times 17 + 3) = 11 \times 88$$

The number has factors other than 1 and itself (for example 11 and 88).

Hence it is a composite number.

3 Irrational numbers and their proofs

4. Prove that $5 - 2\sqrt{3}$ is an irrational number. It is given that $\sqrt{3}$ is an irrational number.

CBSE 2024, 2 marks

SOLUTION

Assume, to the contrary, that $5 - 2\sqrt{3}$ is rational. Then $5 - 2\sqrt{3} = \frac{a}{b}$, where a and b are integers and $b \neq 0$.

Rearranging: $\sqrt{3} = \frac{5b - a}{2b}$

Since a and b are integers, the right side is rational, so $\sqrt{3}$ is rational. This contradicts the given fact that $\sqrt{3}$ is irrational.

Hence $5 - 2\sqrt{3}$ is irrational.